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Information, emotion, and direct democracy: evidence from a survey experiment on Taiwan's nuclear power plant referendum

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ABSTRACT

Information and cue-taking research suggests that given most citizens lack knowledge of and interest in politics, people rely on cues from political parties or social groups to decide their opinions on issues. Using a survey experiment conducted in Taiwan regarding nuclear energy, we rigorously examine the effect of information on direct democracy. Further, based on appraisal theories I suppose that information as stimuli may trigger emotion if there is significant policy inference from information. This study estimates a causal mediation effect of information on voting intention through emotion. The central finding is that receiving more relevant information triggers emotion, which in turn makes individuals in such circumstance much more likely to express their voice (i.e. participate in the referendum). [This study echoes] the information research: party-cue or group-influence is not a necessary condition for people to accept new information and behave accordingly.

KEYWORDS

Mediation analysis;
referendum; Taiwan; nuclear
power; emotion

Introduction

In recent decades, scholars have increasingly focused on the impact of party or group cues on public opinion. This research suggests voters often rely on cues from political parties or social groups to form their opinions instead of personally delving into the specifics of an issue. Much of the research has concentrated on cue-based information processing, and there has been limited discussion on the role of the information itself in shaping attitudes. This focus may lead to an overestimation of the influence of political parties on policy attitudes. However, some studies have demonstrated that individuals consider factual information when forming opinions regardless of party cues. Both lines of research have significant policy implications, but most conclusions have been drawn from counterfactual experiments and are limited to Western countries. This study investigates whether these findings are universally applicable or specific to particular contexts.

Our research builds on existing theories by examining how Taiwanese citizens respond to emotions elicited by information when making voting decisions in referendums. Taiwan is a good case for this analysis as it is recognised for its

vibrant democratic system, and direct democracy has been a distinctive feature since the passage of the Referendum Act by the Legislative Yuan in 2004. Significant revision to the Referendum Act in 2016 significantly lowered the threshold for proposing and approving referendums to 25% of the eligible voters. As a result, numerous proposals seeking to change important public policies emerged. Some of these proposals successfully met the quorum in 2018, prompting the ruling party, the Democratic Progressive Party (DPP), to amend the Referendum Act once again in 2019. One of the changes to the Referendum Act splits the day that national referendums are held from the day that elections are held. With referendums no longer occurring alongside polls, voter turnout is likely to decrease, making it less probable that referendums will meet the necessary benchmarks to be binding.

Prior to 2016, many referendums in Taiwan were heavily influenced by partisan politics, such as the issue of government procurement of additional anti-missile systems. Furthermore, for a referendum to be considered valid, a quorum requiring at least 50% of eligible voters to vote 'yes' was in place. To defeat a referendum, the opposition party encouraged its supporters not to participate. Consequently, national identity and party politics had a significant impact on Taiwan's direct democracy.¹ Voter preferences regarding referendums were closely linked to partisanship, national identity, and cross-Strait relations, as highlighted in previous studies (Bedford and Hwang 2006; Huang 2008; Kao 2004; Mattlin 2004). However, there has been limited focus among Taiwanese scholars on the role of information in influencing vote choices in referendums, unlike studies conducted in other countries (e.g. Lin 2022; Lupia and Matsuzaka 2004).

Through a rigorous experimental design, this study explores a relatively under-analysed topic: the impact of information on direct democracy. We analyse data obtained from an original survey experiment conducted between 24 February 2021, and 11 March 2021. The study took place approximately six months ahead of the referendums scheduled for 18 December 2021. The primary objective is to investigate the causal effect of relevant information on attitudes. The key finding suggests that when individuals receive more relevant information, it triggers emotions, which significantly increases their likelihood of participating in the referendum and expressing their views. Specifically, this research estimates the causal mediation effect of information on turnout intention through emotions. By demonstrating that significant facts can greatly enhance political participation, this study highlights the importance of information, challenging the notion that party cues or group influence are necessary for individuals to accept new information and act accordingly.

This research specifically investigates whether individuals' turnout intention and vote choice are influenced by the information they possess regarding an energy policy. The results of this study have at least one normative and one empirical implication. From the normative perspective, our findings suggest that individuals' attitudes can be affected by information without party cues. While Druckman, Peterson, and Slothuus (2013) point out that under party polarisation, party cues exert greater influence on decision-making, our evidence implies that citizens are

still likely to derive their issue positions from policy information without partisan endorsement, even in the context of referendums.

Secondly, from an empirical standpoint, the findings demonstrate that relevant information can impact political behaviour by eliciting emotional responses towards the policy. While national identity continues to have a significant impact on various aspects of society, our findings suggest that information and emotions are increasingly influential in politics. This does not imply that the level of information or sophistication alone determines political attitudes, but rather emphasises the need to pay greater attention to information and emotions in political analysis (Boudreau and MacKenzie 2014).

The subsequent sections provide a literature review on information processing and the use of party cues, formulate hypotheses based on previous research, employ a causal mediation model to estimate the mechanism's effect, and conclude with a discussion of the findings.

Information, affect, and behavior

The concept of 'low-information rationality' indicates that people can make rational decisions with limited information (Popkin 1991). It is suggested that the general public often struggles to engage with the intricacies of important issues (Delli Alvarez 1998; Carpini and Keeter 1996), but individuals still make reasoned choices through the utilisation of cognitive heuristics (Lupia and McCubbins 1998). Additionally, there is abundant literature supporting the theory of 'cue-taking'. Political parties (Arceneaux, 2008; Campbell et al. 1960; Green, Palmquist, and Schickler 2008; Rahn, 1993), elites (Brader, Valentino, and Suhay 2008; Bullock 2011; Zaller 1992), politicians (Nicholson 2011, 2012), and social groups (Brady and Sniderman, 1985; Cohen 2003; Gilens 1999; Lupia 1994) provide cues or signals that individuals use to form opinions or make judgements about political candidates, policies, or other social phenomena. This approach provides insights into how individuals navigate complex information environments and use cues to simplify decision-making processes in political and social settings.

Extensive research has explored cognition as an independent means of assessing the state of the world (Robert, Goodin, and Spiekermann 2018). Relying on cues rather than seeking detailed information can be a rational decision-making strategy. The theory of 'cue-taking' investigates how individuals can make decisions, such as voting, even with incomplete information. Conversely, the selective exposure theory suggests that people tend to select information that aligns with their existing beliefs (Arceneaux and Johnson 2013; Knobloch-Westerwick and Meng 2009; Stroud 2018). Thus, it is challenging to disentangle the overlapping influences of party cues and pre-existing beliefs. Consequently, previous research may overestimate the impact of cue-taking on attitudes. Additionally, social media algorithms may filter out opposing viewpoints, potentially limiting the external validity of experiments that emphasise the effects of cue-taking.

While acknowledging that the cue-taking theory provides insights as to how people use cues to simplify decision-making in a complex information environment, scholars emphasise the importance of affect (Boudreau and MacKenzie 2014;

Kuklinski and Hurley 1994). The tripartite model of attitudes connects cognitive, affective, and behavioural components (Eagly and Chaiken 1993). Affect refers to emotions, fear, and anxiety, while cognition involves thoughts, beliefs, and inferences (Brader 2006). While information plays a crucial role in shaping preferences and actions, affect, or emotions, can serve as a mediator between information and behaviour. Affect also bears on pleasurable or unpleasurable emotions or feelings towards an object (Johnson and Boynton 2010). Emotions can be categorised as positive or negative, with negative emotions being more likely to arise under threatening conditions (Huddy 2007). Affective intelligence theory suggests that affect, either alone or in interaction with other variables, can enhance individuals' interest in specific subjects (Brader 2006). Thus, this paper posits that information influences attitudes through specific emotions.

In the realm of political psychology, there is an extensive body of research examining how information can elicit affect and, consequently, influence behaviour. For example, appraisal theories propose that specific emotions are evoked through an evaluation or appraisal process, whether conscious or unconscious, of people, communications, events, or life situations (Justin, Crigler, and Belt 2007, 232). According to Brader and Marcus (2013, 167), appraisals can be seen as cognitive interpretations of the significance of a situation for one's goals. Building upon appraisal theories, this research hypothesises that information, acting as stimuli, may trigger emotions if there are significant policy implications inferred from that information.

An expanding body of research delves into the influence of affect on political actions and attitudes. Sniderman, Hagendoorn, and Prior (2004) proposed that perceptions of social groups can evoke emotions related to potential threats from another group. Brader, Valentino, and Suhay (2008) argued that emotions towards immigrants can be shaped by elite discourses containing cues related to ethnicity or race, emphasising the emotional impact of such discourses. Kuklinski and Hurley (1994) incorporated the affect factor into their research by assessing individuals' affinity for elites who comment on racial issues, highlighting the affective basis of cue-taking. Forgas (2008) demonstrated that negative affect can lead to avoidant or defensive behaviour. Bullock's (2011) experiments revealed that individuals' inclination to engage in information-processing activities moderates the effects of policies and elite influence. It is not a given that individuals will solely respond to party cues when presented with policy content. Boudreau and MacKenzie (2014) examined how policy information and party cues jointly influence public opinion, finding that individuals may respond to policy information even when exposed to party cues.

In this study, we examine the influence of negative emotions on action intentions using the nuclear power plant referendum as a case study. Our focus is on how engaging information about the energy source can elicit specific emotions, which subsequently impact turnout intention and vote choice. Given Taiwan's lack of natural resources and the government's emphasis on economic development, the nuclear power issue is highly salient and potentially controversial for some individuals. The policy implications of this study are significant, as the outcome variable is the intention to vote in a referendum. Before presenting our hypotheses

and research design, we provide a brief overview of the background of the nuclear power plant and the referendum.

Taiwan's energy policy and referendum

In order to meet its domestic energy demands, Taiwan relies on imports of coal, gas, and oil for thermal power generation. To ensure a stable energy supply, Taiwan began building nuclear power plants in the 1970s. After establishing two nuclear power stations in northern Taiwan and one in the south, the government decided to proceed with the construction of a fourth plant in northern Taiwan in 1980. However, this project encountered significant opposition from local residents and scholars. One of their primary concerns was the safety of nuclear power plants, especially given their location in densely populated areas like northern Taiwan. The frequent occurrence of earthquakes in Taiwan has intensified attention to this safety issue. Additionally, concerns were voiced regarding the disposal of nuclear waste, which contains radioactive materials. Most of the waste is currently stored within the nuclear power plants, while a portion has been transported to a remote island.

The Democratic Progressive Party (DPP) has been supportive of the anti-nuclear power movement since the 1980s (Ho 2003), and has actively mobilised people through street protests and sit-ins to garner support for the cause. Despite years of debates and changes in political party leadership in 2000 and 2008, no consensus was reached among political parties regarding the construction of the Fourth Nuclear Power Plant. On 11 March 2011, a major earthquake occurred 130 kilometres off the coast of Japan, leading to a failure in the cooling system of three reactors at the Fukushima power plant. This accident, which was televised worldwide, caused a global shift in energy policy. For example, Germany decided to shut down nuclear energy completely by 2022, with the immediate closure of eight of its oldest nuclear power plants. In Taiwan, public concerns over the existing and proposed nuclear power plants also intensified.

In response to public opposition following the Fukushima accident, the Kuomintang (KMT) government announced the decommissioning of the Fourth Nuclear Power Plant on 28 April 2014, reflecting the low acceptance of nuclear energy among the public. Despite this, the three existing nuclear power plants continue to operate, generating approximately 18% of the country's electricity at that time. When the DPP won the presidential election in 2016, the new administration pledged to replace all nuclear power with solar and wind power, aiming for a 'nuclear-free homeland' by 2025. This commitment was subsequently incorporated into the Electricity Act. Proponents of nuclear energy argue that it is a cleaner and safer alternative to coal and gas-fired energy, and they believe that the issue of radioactive waste can be resolved in the near future. In 2018, these advocates successfully proposed a referendum to remove the 'nuclear-free homeland' clause from the Electricity Act,² which was approved by 54.42% of votes.³ Although the clause was removed from the Electricity Act following the referendum, the DPP government did not abandon the 'nuclear-free homeland' policy. This decision disappointed supporters of nuclear energy, who proposed another referendum to

revive the construction of the Fourth Nuclear Power Plant.⁴ Meanwhile, there were additional proposals on food security, algae reef protection, and referendum dates.⁵ The four referendums were scheduled for 28 August 2021, but they are postponed to 18 December 2021 due to the pandemic.

Despite frequently promoting the target of renewable energy comprising 20% of total electricity, then president Tsai Ing-wen and premier Su Tseng-chang often overlooked the significant fact that green power constituted only 6% of electricity generation. Throughout Tsai's first term, gas-fired power remained the primary source of electricity generation, followed by coal-fired power.⁶ According to the Taipower Company, the share of renewable energy, which includes solar, wind, and geothermal power, was 5.1% in 2016. This figure rose to 6% by 2019.⁷

To maintain fairness, it should be noted that the DPP government has not made any false claims about renewable energy becoming the major source of electricity in the near future. In fact, the three nuclear power plants only generate around 10% of the country's electricity after 2017, which is slightly more than wind and solar power combined. However, the DPP has government consistently emphasised the potential and progress of green power while providing limited information about the existing disparity between the current state—6% of electricity from renewable sources – and the future goal of reaching 20% by 2025.

On 15 August 2017, a nationwide blackout occurred when a thermal power plant in Taoyuan malfunctioned, resulting in a five-hour power outage that affected around six million households. The DPP government faced significant criticism for moving away from nuclear power before having sufficient green energy alternatives in place. The Minister of Economic Affairs took responsibility for the incident and resigned. While the government attributed the incident to issues with the operation of natural gas at the power plant, the opposition party, media, and foreign investors expressed concerns about Taiwan's energy policy, particularly its heavy reliance on renewable energy and thermal power.

As a result of the COVID-19 pandemic, the nuclear power plant referendum, originally scheduled for August 2021, was rescheduled to 18 December 2021. According to the Referendum Act, a referendum must be approved by 25% of eligible voters to pass. While the KMT and supporters of the nuclear power plant campaigned extensively to encourage citizens over the age of 18 to vote yes, the DPP government urged people to vote no by emphasising the risks of a nuclear power plant and its radioactive waste. Ultimately, the nuclear power plant referendum, along with the other three referendums held on the same day, was rejected because it did not meet the 25% quorum of approval among all eligible voters. The total turnout for the referendum was 41%, with only 47% of valid votes, or 19% of the eligible electorate, in favour of the initiative. This research examines the role of information and affect in both turnout intention and vote choice in this referendum.

Research hypotheses

Drawing from previous studies, this research proposes that exposure to policy information can influence individuals' attitudes towards policies. The first

hypothesis posits that informing people about the proportion of renewable energy will elicit responses to the policy related to this information. Specifically, individuals who learn that renewable energy constitutes only 6% of electricity generation are more likely to develop negative perceptions about the overall energy policy compared to those who do not receive this information. Since there is no specific source cited for the policy information, respondents must evaluate its credibility on their own, without relying on party or group cues. If the results of this experiment demonstrate differences in responses based on the information provided in the survey, it suggests that the information gap itself can significantly affect policy attitudes.

According to Brader, Valentino, and Suhay (2008), emotions towards a policy become more intense when individuals receive information that opposes the policy goal. In this study, it is expected that incoming policy information will elicit concerns regarding energy shortages. Consequently, heightened worry is likely to increase the intention to express opposition to the policy. Hence, the second hypothesis posits that receiving negative policy information triggers worry regarding the energy policy, thereby increasing the likelihood of participating in and voting in favour of the referendum proposal on restarting nuclear power plant construction.

To address model specification concerns, the study incorporates controls for the general evaluation of renewable power in both the mediator and outcome models. Respondents are asked to evaluate the positive or negative impact of the government's promotion of wind and solar power on the country.

Our hypotheses can be summarised in the following two diagrams:

Experiment group: Policy information → (+) Worry → (+) Participation in the referendum

Control group: Absence of policy information → (x) Worry → (x) Absent in the referendum

We propose the following hypotheses to examine the causal relationships between information and the level of worry, and between the level of worry and intention to turn out in the referendum.

H1: Policy information increases the level of worry about energy.

H2a: The level of worry about energy increases the intention to turn out in the nuclear power plant referendum.

Regarding the role of party identification in the turnout intention model, it is assumed that both Pan-Green and Pan-Blue⁸ supporters are mobilised by their respective parties (DPP and KMT) to participate in the referendum. Pro-nuclear civic groups, with the assistance of the KMT, initiated the referendum petition, while the DPP has long advocated for a no-nuclear power stance. Consequently, the parties campaigned to rally their supporters to vote 'yes' and 'no' in the referendum, respectively.

To confirm the influence of emotion elicited by information, we assess individuals' overall evaluation of the energy policy and incorporate this variable into the model. If

policy evaluation can counteract emotion, both affect and information become irrelevant to political behaviour. Under this scenario, individuals rely solely on their cognition to shape their attitudes and determine their political participation. However, given that emotion is influenced by the information provided in this study, we anticipate it will exert a stronger influence than cognition on attitudes towards the energy policy. Therefore, emotion is expected to predominate in both the mediator and outcome models, even when general evaluations are considered in both models.

Turning to voter choice, individuals who receive policy information may develop emotions concerning the energy issue, leading them to vote in favour of the nuclear power plant referendum (voting 'yes'). Similarly, it is assumed that Pan-Blue supporters are more inclined to vote 'yes' to support restarting construction of the Fourth Nuclear Power Plant, while Pan-Green identifiers tend to vote 'no' regardless of the presence of policy information.

Experimental group: Policy information → (+) Worry → (+) Approve the proposal for restarting construction of the Fourth Nuclear Power Plant

Control group: Absence of policy information → (x) Worry → (x) Approve the proposal for restarting construction of the Fourth Nuclear Power Plant

We propose the following hypothesis to examine the causal relationship between the level of worry and intention to vote in favour of the referendum.

H2b: The level of worry about energy increases the intention to vote in favour of the nuclear power plant referendum.

Survey experiment

The survey data were collected through household personal communication conducted between January 24 through 11 March 2021. Participants eligible for this survey were individuals over 18 years old with household registration in Taiwan. Telephone numbers were systematically sampled from telephone directories. In order to reach households with unlisted numbers, the last two digits of the selected telephone numbers were randomised. The sample was randomly divided into two groups: the experimental group and the control group. The experimental group had 1,157 completed interviews, while the control group had 1,166 completed interviews. According to the AAPOR response rate calculator, the response rates were 7.3% and 8.8%, respectively.⁹ Overall, there were 2,323 successful interviews out of a total of 21,242 calls, resulting in a response rate of 7.03%. The results for the full sample have a margin of sampling error of plus or minus 2.07%.

In this survey, respondents were asked about their likelihood of voting in the upcoming August referendum and, if they intended to vote, whether they would vote 'yes' or 'no'. Out of 2,323 respondents, 1,428 (equivalent to 71.47%) indicated that they were likely or very likely to vote, while 586 (28.52%) stated they would not vote. Additionally, 269 respondents chose not to answer this question. Among the 1,428 likely voters, 753 (52.73%) expressed a preference for 'yes', 384 (26.89%) favoured 'no', and 291 (20.37%) were undecided or declined to reveal their decision. Excluding the undecided voters, the survey suggested that 66.23% of committed voters would vote in favour the referendum.

However, the actual referendum result showed 47.16% in favour of 'yes' and 52.84% for 'no'. This disparity suggests that the time between the survey and the referendum likely contributed to overestimating both the voter turnout rate and the approval rate.

This research was overseen by the Institutional Review Board (IRB) at National Chengchi University. An IRB-approved consent protocol was used to obtain voluntary and informed consent from participants, with no deception involved. Prior to beginning the survey, informed consent was obtained verbally from all respondents via telephone. Interviewers used a consent script that was programmed into the computer software, and respondents had to consent in order to proceed to the interview.

The experimental group received the following question: 'The government has recently promoted wind power and solar power. However, green power currently constitutes only 6% of total electricity generation. Are you worried that the government's approach will lead to a shortage of electricity?' The control group received a slightly different question: 'The government has recently promoted wind power and solar power. Are you worried that the government's approach will lead to a shortage of electricity?'

Both groups were informed about the government's promotion of green power. However, the experimental group received additional information stating that green power currently accounts for only 6% of total electricity generation. This information is expected to trigger anxiety among respondents regarding Taiwan's future energy supply, as a single-digit proportion of energy does not sound very promising to the general public. Therefore, we hypothesise that the experimental group will exhibit a higher level of worry, and this level of worry will be positively associated with the probability of voter turnout.

During the survey, respondents were also asked about their political affiliation and demographic background. Those identifying with the KMT, New Party, and People's First Party were coded as Pan-Blue, while those identifying with the DPP, New Power Party, and Taiwan Statebuilding Party were coded as Pan-Green. It is anticipated that Pan-Blue partisans are more likely to vote 'yes' in the pro-nuclear referendum, while Pan-Green identifiers are more inclined to either abstain or vote 'no'. Demographic information collected includes gender, age, education, occupation, Taiwanese identity, and area of residence. For more detailed information on the wording of the questions, please refer to the [Appendix](#).

Tingly et al. (2013) developed a causal mediation model that is accompanied by the R package mediation. Unlike traditional structural equation models (SEM), their causal mediation model employs a counterfactual approach to estimate causal inference under certain assumptions. This counterfactual approach, which requires randomisation of the treatment, ensures that no omitted variables, such as demographic variables and attitudes, influence the relationship between the independent and dependent variables. More importantly, this method can avoid confounding between the mediator and dependent variable (García, Seco, and Lozano 2014).

Following Tingly et al. (2013), the total causal effect can be decomposed into direct and indirect effects, as shown in equation (1):

$$\tau_i = \delta_i(t) + \zeta_i(1 - t) \quad (1)$$

$\delta_i(t)$ represents the difference between outcome Y when the mediator M is present versus when it is absent. $\zeta_i(t)$ represents the difference between outcome Y when the treatment t is administered versus when it is not administered. Therefore, τ_i represents the difference between outcome Y when the treatment t is administered or not administered, and mediator M is present or absent. When treatment t is administered, $\tau_i = \delta_i(1)$, indicating the presence of a mediation effect. Conversely, when the treatment is not administered, $\tau_i = \zeta_i(1)$, reflecting the effect of the treatment.

Tingly et al. (2013) introduced the concept of average causal mediation effect (ACME) and average direct effect (ADE), which represent the population-level effects of mediation and direct pathways, respectively. In this study, we employ two statistical models to estimate these effects: the mediator model and the outcome model. The mediator model estimates the conditional distribution of the mediator M_i given the treatment T_i and a set of the observed pre-treatment covariates X_i (equation 2). The outcome model estimates the outcome Y_i given T_i , M_i , and X_i (equation 3). These two models are fitted separately, and their fitted objects are subsequently analysed using the mediate function to compute the estimated ACME and other quantities of interest.

$$M_i = \alpha T_i + \beta X_i \quad (2)$$

$$Y_i = \gamma M_i + \alpha T_i + \beta X_i \quad (3)$$

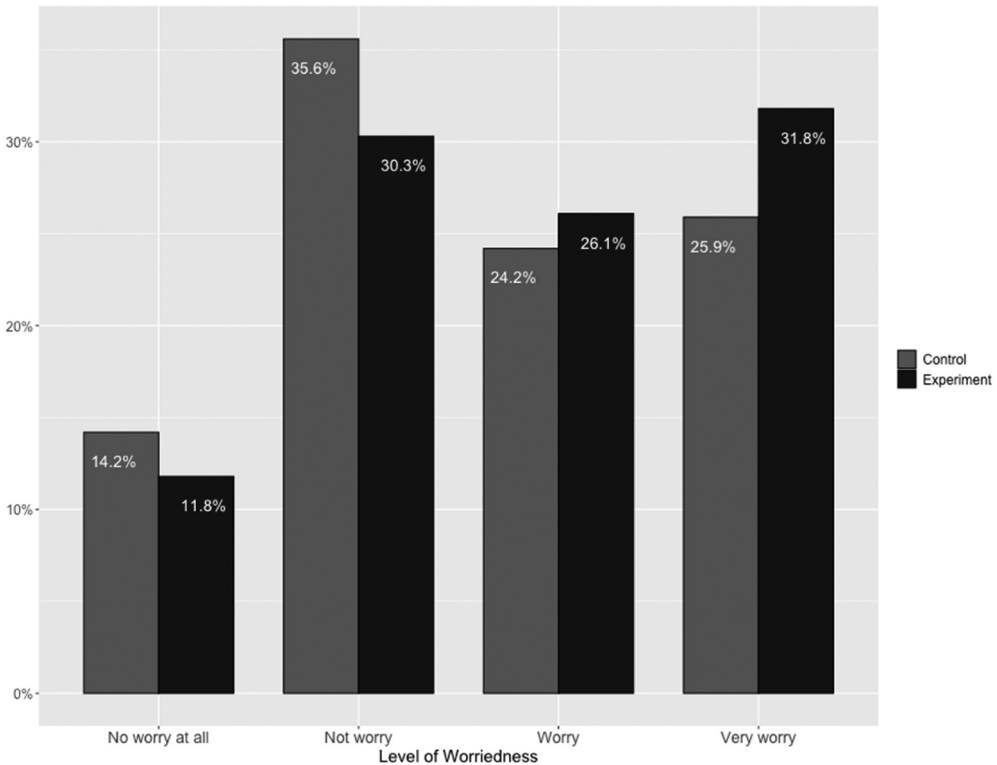


Figure 1. Level of worry of control and experiment groups. Average level of worry of the experiment and control group is 2.77 (s.d. = 1.02) and 2.61 (s.d. = 1.01). t value = 3.63, p -value < 0.001.

Table 1. Estimates of worry model as mediator for voting intention.

	D.V. = worry Not worried at all = 1, Very worried = 4	
	Coef.	S.E.
(Intercept)	1.709***	0.191
Treatment	0.218***	0.043
Negative evaluation on green energy policy	0.424***	0.028
Pan-Blue	0.166**	0.033
Pan-Green	-0.340***	0.053
Taiwanese identity	-0.207***	0.052
Female	0.023	0.053
Age (base: 20-29 years old)		
30-39 years old	-0.126	0.086
40-49 years old	0.021	0.080
50-59 years old	0.193*	0.080
60+ years old	0.207**	0.081
Education(base: Elementary school)		
Junior high school	-0.107	0.123
Senior high school	0.046	0.107
Some college	0.117	0.116
University	0.179	0.112
Residence area(base: Hualien and Taitung)		
Taipei metropolitan area	-0.049	0.130
New Taipei and Keelung	-0.030	0.145
Tayuan, Hsinchu, and Miaoli	0.093	0.132
Taichung, Chungwa, and Nantou	-0.041	0.127
Yunlin, Chiayi, and Tainan	0.020	0.128
Kaohsiung, Pingtung, and Penghu	-0.103	0.131
Occupation(base: blue collar)		
White collar (managers)	-0.083	0.057
While collar (office workers)	-0.029	0.063
Farmers and Fishermen	-0.079	0.111
N	1663	
Adjusted R ²	0.277	
F-Statistics	28.74***	

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, $\$ < 0.1$.

Experiment results

There are two types of causal mediation analysis: turnout intention and vote choice. Figure 1 shows the levels in both the control and experimental groups. The experimental group exhibits higher levels of emotion and lower absence of emotion compared to the control group. Participants exposed to the '6% message' expressed greater worry about future energy issues.

Table 1 presents the results of the mediator model for turnout intention, with the dependent variable being the level of worry measured on a 4-point scale. The findings indicate that receiving the treatment, which involves providing information about the share of green power, increases worry levels by 0.218 points on average. This effect holds true even after controlling for factors such as general evaluation of green power policy, party identification, gender, age, and education. The empirical evidence supports $H1$. Additionally, a one-unit increase in negative evaluation of the green power policy is associated with a 0.424 point increase in the level of worry. Among respondents, Pan-Blue supporters are more worried about Taiwan's energy policy than other individuals by 0.166 points on average. Conversely, Pan-Green supporters and Taiwanese identifiers are

Table 2. Estimates of voting intention model.

	D.V. = voting intention turnout = 1, abstention = 0	
	Coef.	S.E.
(Intercept)	−0.209	0.526
Worry	0.397***	0.066
Negative evaluation on green energy policy	0.148	0.084
Treatment	0.087	0.117
Pan-Blue	0.034	0.100
Pan-Green	0.245	0.139
Taiwanese identity	−0.270	0.138
Female	−0.151	0.144
Age (base: 20-29 years old)		
30-39 years old	−0.240	0.233
40-49 years old	−0.019	0.220
50-59 years old	−0.285	0.220
60+ years old	−0.07	0.224
Education(base: Elementary school)		
Junior high school	0.104	0.297
Senior high school	0.233	0.261
Some college	−0.005	0.284
University	0.841*	0.281
Residence area(base: Hualien and Taitung)		
Taipei metropolitan area	−0.448	0.368
New Taipei and Keelung	−0.735	0.397
Taoyuan, Hsinchu, and Miaoli	−0.110	0.379
Taichung, Chungwa, and Nantou	−0.321	0.362
Yunlin, Chiayi, and Tainan	−0.450	0.361
Kaohsiung, Pingtung, and Penghu	−0.399	0.370
Occupation(base: blue collar)		
White collar (managers)	0.215	0.148
White collar (office workers)	0.172	0.166
Farmers and Fishermen	0.031	0.276
N	1663	
−2 log-likelihood	−901.98	
McFadden's R ²	0.071	

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

less worried about the energy policy by 0.340 and 0.207 points on average, respectively. Moreover, older individuals tend to be more worried about the energy policy.

Table 2 presents the estimates of the outcome model, where the dependent variable is respondents' likelihood of participating in the referendum. Respondents who indicated they would definitely or probably vote are coded as 1, while those who stated they probably would not vote or definitely would not vote are coded as 0. Since it is a dichotomous variable, a binary logistic regression model is employed to estimate the effects of the independent variables.

A key observation from Table 2 is that the level of worry significantly influences turnout intention. Individuals who express worry about energy issues are more inclined to participate in the nuclear power plant referendum to voice their opinions. Notably, the level of worry is strongly correlated to whether respondents have received the information that renewable energy accounts for only 6% of electricity generation. Consequently, this information affects turnout intention through emotional responses. This finding confirms *H1a*. Additionally, Table 2 indicates that highly educated voters are more likely to participate in the referendum.

Table 3. Estimates of mediation effect and nonparametric bootstrap confidence intervals of voting intention model.

	Estimate	95% C.I. Lower	95% C.I. Higher
ACME (control)	0.015***	0.008	0.02
ACME (treated)	0.015***	0.008	0.02
ADE (control)	0.016	−0.023	0.06
ADE (treated)	0.015	−0.022	0.06
Total effect	0.031	−0.005	0.07
ACME (average)	0.015***	0.008	0.02
ADE (average)	0.015	−0.022	0.06
Prop. Mediated Effect (average)	0.450	−1.762	2.70
N	1663		
Simulations	500		

*** $p < 0.001$.

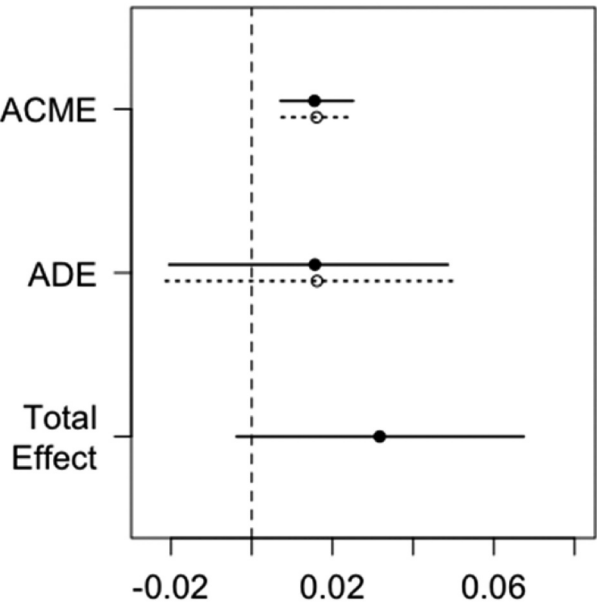


Figure 2. ACME, ADE, and total effect from the mediation function of turnout. Note: Solid line represents the effect of treated group, and dashed line represents the effect of control group.

Following the estimation of the mediator and outcome models, we utilised the mediation function to estimate the average causal mediation effect (ACME) and average direct effect (ADE). ACME represents the indirect effect of the information on the likelihood of turnout through the pathway of worry. Table 3 presents the estimates and 95% confidence intervals, while Figure 2 illustrates the results of the mediation function. The estimated ACME of 0.015 is statistically significant, indicating a meaningful indirect effect. The proportion of the mediated effect is 0.45, suggesting that nearly half of the total effect of the treatment on the outcome is attributed to the mediator. However, the estimated average direct and total effects are not statistically significant. Therefore, there is robust statistical evidence supporting the presence of mediation in the data. These findings

suggest that the treatment provided in the experiment increases worry levels, which in turn motivates individuals to participate in the referendum.

The initial causal mediation analysis confirms the causal impact of information on turnout intention, revealing that informed individuals express greater worry about energy issues. Consequently, information has the potential to enhance turnout intention by stimulating worry. Importantly, even when respondents are asked to consider the overall benefit of green power to Taiwan, the information still triggers worry, which in turn motivates them to vote in the referendum. In other words, the causal effect of information on turnout is not merely coincidental.

Considering that both political camps aim to mobilise voters, it is crucial to ascertain whether information truly influences voting decisions in the referendum. Similar to [Tables 1, 4](#) presents the estimates of the mediator model for the 4-point level of worry. Receiving information about green power accounting for 6% of electricity leads to an average increase in the level of worry of 0.121 points. Additionally, a one-unit increase in the negative evaluation of the green power policy causes an average increase in the level of worry of 0.389 points. Pan-Blue supporters exhibit greater levels of worry compared to others by an average of 0.165 points, while Pan-Green supporters and Taiwanese identifiers express less worry by an average of 0.497 and 0.226 points, respectively.

[Table 5](#) reveals a statistically significant association between the level of worry and voting in favour of the referendum. Specifically, a one-unit increase in the level of worry corresponds to a 0.750 increase in $\log(p/1-p)$, which translates into a 111% ($\exp(0.750)-1$) increase in the odds of approving the referendum, assuming that all other variables remain constant. Additionally, a one-unit increase in the negative assessment of renewable energy results in a 68% increase in the odds of approving the referendum ($\exp(0.519)-1$). The odds of voting

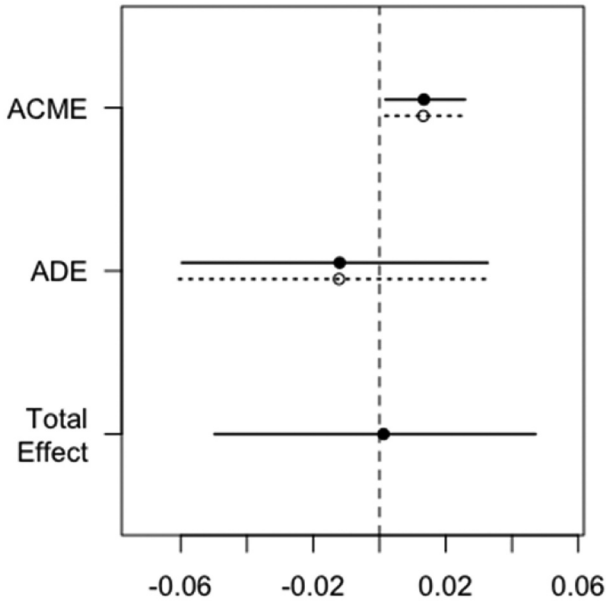


Figure 3. ACME, ADE, and total effect from the mediation function of vote yes. Note: Solid line represents the effect of treated group, and dashed line represents the effect of control group.

Table 4. Estimates of worry model as mediator for voting.

	D.V. = worry	
	Not worried at all = 1, Very worried = 4	
	Coef.	S.E.
(Intercept)	2.020***	0.260
Treatment	0.121*	0.057
Negative evaluation on green energy policy	0.389***	0.034
Pan-Blue	0.165**	0.044
Pan-Green	-0.407***	0.071
Taiwanese identity	-0.226**	0.069
Female	-0.051	0.070
Age (base: 20-29 years old)		
30-39 years old	-0.218	0.109
40-49 years old	-0.068	0.103
50-59 years old	0.132	0.104
60+ years old	0.186	0.104
Education(base: Elementary school)		
Junior high school	-0.033	0.175
Senior high school	-0.017	0.154
Some college	0.154	0.168
University	0.146	0.158
Residence area(base: Hualien and Taitung)		
Taipei metropolitan area	0.046	0.169
New Taipei and Keelung	0.050	0.191
Taoyuan, Hsinchu, and Miaoli	0.126	0.168
Taichung, Chungwa, and Nantou	0.015	0.162
Yunlin, Chiayi, and Tainan	0.120	0.164
Kaohsiung, Pingtung, and Penghu	0.015	0.168
Occupation(base: blue collar)		
White collar (managers)	-0.081	0.076
While collar (office workers)	-0.071	0.084
Farmers and Fishermen	-0.262	0.160
N	1663	
Adjusted R ²	0.277	
F-Statistics	28.74***	

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

in favour are predicted to be about 1.67 ($\exp(0.514)$) times larger among Pan-Blue supporters compared to other individuals. Conversely, the odds of voting in favour are about 0.336 and 0.519 times lower among Pan-Green supporters and Taiwanese identifiers, respectively. These findings support the hypothesis that emotional responses triggered by information can influence voting behaviour according to the information provided to individuals. Regarding demographic factors, individuals with a junior high school education are more likely to vote against the nuclear power plant referendum, while those whose livelihoods depend on farming or fishing are more likely to support it.

Table 6 presents the estimated ACME and ADE of the level of worry as the mediation variable. The findings indicate that the ACME is statistically significant at the 95% confidence level. The proportion of the mediated effect is 0.187, indicating that the mediator accounts for about 20% of the total effect. Figure 3 visually represents the results of the mediation function, illustrating the ACME for both the control and experimental groups. The ACME values, 0.013 for both groups, are

Table 5. Estimates of voting model.

	D.V. = voting intention turnout = 1, abstention = 0	
	Coef.	S.E.
(Intercept)	-1.372***	0.768
Worry	0.750**	0.093
Negative evaluation on green energy policy	0.519**	0.119
Treatment	-0.079	0.170
Pan-Blue	0.514**	0.165
Pan-Green	-1.088***	0.191
Taiwanese identity	-0.655**	0.197
Female	-0.263	0.211
Age (base: 20-29 years old)		
30-39 years old	0.262	0.300
40-49 years old	-0.274	0.286
50-59 years old	-0.221	0.297
60+ years old	-0.398	0.299
Education(base: Elementary school)		
Junior high school	-1.188*	0.504
Senior high school	-0.498	0.442
Some college	-0.872	0.496
University	-0.661	0.459
Residence area(base: Hualien and Taitung)		
Taipei metropolitan area	0.691	0.499
New Taipei and Keelung	0.287	0.563
Tayuan, Hsinchu, and Miaoli	0.196	0.491
Taichung, Chungwa, and Nantou	0.725	0.474
Yunlin, Chiayi, and Tainan	0.737	0.477
Kaohsiung, Pingtung, and Penghu	0.309	0.491
Occupation(base: blue collar)		
White collar (managers)	0.195	0.220
White collar (office workers)	-0.128	0.239
Farmers and Fishermen	1.258*	0.522
N	980	
-2 log-likelihood	-440.385	
Mcfadden's R ²	0.303	

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.**Table 6.** Estimates of mediation effect and nonparametric bootstrap confidence intervals of voting model.

	Estimate	95% C.I. Lower	95% C.I. Higher
ACME (control)	0.013*	0.002	0.03
ACME (treated)	0.013*	0.002	0.03
ADE (control)	-0.011	-0.058	0.03
ADE (treated)	-0.011	-0.057	0.05
Total effect	0.001	-0.051	0.05
ACME (average)	0.013*	0.002	0.03
ADE (average)	-0.011	-0.058	0.03
Prop. Mediated Effect (average)	0.187	-6.238	9.30
N	980		
Simulations	500		

* $p < 0.05$.

significantly different from zero. These results suggest that the treatment implemented in the information experiment successfully increased the emotional response among respondents. Consequently, they are more inclined to participate in the voting process and vote in favour of the referendum. Despite the relatively small ACME values of 0.015 and 0.013 shown in [Tables 4 and 6](#), respectively, our experiment sheds light on the underlying mechanism linking information exposure with attitudes and subsequent behaviour.

Conclusion

This research shifts the traditional focus from party cues and instead highlights the importance of policy information, which may be less accessible to individuals compared to party or group cues. Additionally, scholars have increasingly acknowledged the role of cognitive ability in understanding policy issues when exposed to relevant information (Boudreau and MacKenzie 2014). Consequently, this study posits that policy information, independent of party cues, can influence both turnout intention and vote choice in a referendum. Through a survey experiment, the study elucidates the underlying mechanism linking information and behaviour, specifically how concerns related to policy implications can be activated by information and subsequently affect voting behaviour in a referendum. The findings indicate that individuals who receive information indicating that green power constitutes only 6% of electricity generation tend to exhibit heightened anxiety about Taiwan's energy situation and a stronger intention to vote in favour of the nuclear power plant referendum.

This study has three important implications. Firstly, since 2004, when former President Chen Shui-bian proposed referenda on 'peace' and 'anti-missile weapons', Taiwan's direct democracy has been perceived more as a strategy for national identity campaigns rather than a means of resolving contentious issues. Following the reduction of the quorum required for approving referendums in 2016, many domestic issues have transformed into referendum issues. A central issue pertains to whether Taiwan should pursue increased nuclear power generation or move towards eliminating nuclear power. Instead of focusing on explaining individual voting choices, this research highlights the previously overlooked importance of information. Evidence from the survey experiment clearly demonstrates that information plays a critical role in influencing support for the pro-nuclear power referendum through emotional channels. Crucially, because policy information is randomly assigned, the relationship between information, worry, and turnout intention and vote choice shows a causal link rather than a mere association.

Despite the absence of references to political parties, elites, or social groups in the information provided to respondents, it still evoked worry about the energy issue. This finding indicates that straightforward policy facts, devoid of party or group cues, can have a more substantial impact on behaviour than initially anticipated. While pure policy information may lack strong ingroup or outgroup positions, it is still crucial for opinion formation. Building upon these findings, future research could explore the effects of various types of policy information on opinion formation. Additionally, it would be valuable to examine individuals' cognitive processes when dealing with complex

information. This approach could provide insights into how governments can effectively persuade people with diverse predispositions and cognitive abilities.

The first limitation of this study is that the treatment provided neutral information, specifically stating that only six percent of electricity is currently generated from renewable sources. This single-digit figure is likely to evoke pessimism among most people, which could explain why the experimental group, on average, displayed greater anxiety about the energy issue compared to the control group. Our mediation analysis highlights that this treatment can influence turnout intention and vote choice through emotional responses, even after accounting for policy evaluation, partisanship, and demographic factors. In the future, incorporating more balanced and realistic information could enhance the robustness of our findings. Another limitation is the discrepancy between the results of the survey experiment and the actual referendum. When respondents were asked about their turnout intention and vote choice, many may not have finalised their decisions. Reducing the time lag between the survey and the referendum could potentially provide a better understanding of respondents' voting behaviour.

Numerous studies suggest that political parties provide biased information or cues to citizens, as not all individuals seek fair and impartial information from political parties. Instead, people often rely on cues or heuristics from their ingroup or outgroup to decide their positions on issues. This poses a challenge to governance and reinforces political polarisation. The present research aims to validate previous scholarly findings that emphasise the cognitive ability of individuals when policy information is presented (Key 1966; Nie, Verba, and Petrocik 1976). On certain issues, the effect of information and values on public opinion is contingent on individual differences in underlying predispositions (Alvarez and Brehm 2002). In other words, reliance on cue-taking is not ubiquitous. Voting based on accurate information may incentivise political parties to move away from negative campaigns or disinformation. Consequently, a critical question for future research is when and how citizens actively seek out policy facts in the context of increasing partisan polarisation.

Notes

1. In Taiwan, the issue of national identity has been prominent since the country's democratisation in the 1990s. National identity refers to 'an individual's psychological connection to a political community' (Wang and Liu 2004, 570). Chinese nationalism, rooted in culture, ethnic ties, and authoritarian governance, has competed with Taiwanese nationalism, which opposes reunification and advocates for a distinct Taiwanese culture. While Taiwanese identity suggests independence from China, many who identify as Taiwanese would support Taiwan's independence only under specific circumstances (Hsieh and Niou 2005; Niou 2004). Therefore, we use the term 'national identity' to denote a collective affiliation without immediate political implications.
2. The referendum question asked: Do you agree to repeal Article 95 paragraph 1 of the Electricity Act: The nuclear energy-based power-generating facilities shall wholly stop running by 2025.
3. The revision of the Referendum Act in 2016 stipulated that the results of a referendum are accepted when the tally of affirmative ballots exceeds negative and reaches 25% of registered voters, or approximately 5 million votes.

4. The referendum question asked: Do you agree to the activation of Taiwan's mothballed Fourth Nuclear Power Plant?
5. The three issues to be voted on were whether Taiwan should ban pork imports containing ractopamine, whether referendums should be held on the same day as general elections, and whether a receiving terminal for liquid natural gas should be relocated away from an area of algae reef. We chose to focus on the energy issue because both main parties have been campaigning on the nuclear power issue since the 1990s. Given that the nuclear power debate has persisted longer than the issues of algae reef protection, food security, and date of referendums, it is possible that citizens are influenced by cues from both parties on this issue. Consequently, if our treatment that solar and wind power only account for 6% of electricity generation can affect attitudes and, consequently, behaviour, it is possible to generalise our findings to other issues.
6. <https://www.taipower.com.tw/tc/>.
7. <https://www.taipower.com.tw/tc/page.aspx?mid=212&cid=120&cchk=f3a1b1e0-03e5-45fa-b72e-b28c5cb94f37>
8. The terms 'Pan-Blue' and 'Pan-Green' refer to the political alliances led by the KMT and DPP, respectively. Prior to the party turnover in 2000, some KMT members who opposed the party's shift towards a Taiwanese identity formed the New Party (NP). Following the 2000 presidential election, the People's First Party (PFP) split from the KMT, advocating for cleaner politics. These three parties collectively became known as the Pan-Blue camp. The Pan-Green camp, on the other hand, comprised the DPP and the Taiwan Solidarity Union (TSU), with the TSU actively promoting Taiwan's independence. For further details, please refer to Achen and Wang (2017).
9. The response rate is calculated using the formula: $I/((I+P)+(R+NC+O)+(UH+UO))$, where I is complete interviews, P is partial interviews, R is refusal and break off, NC is non-contact, O is other, UH is unknown household, and UO is unknown respondent eligibility.

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No potential conflict of interest was reported by the author(s).

Notes on contributor

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Appendix

Wording of questions

[Treatment: Worry]

Q16-a. The government has recently promoted wind power and solar power. However, green power currently constitutes only 6 percent of total electricity generation. Are you worried that the government's approach will lead to a shortage of electricity?

(Very worried, worried, not too worried, not at all worried)

Q16-b. The government has recently promoted wind power and solar power. Are you worried that the government's approach will lead to a shortage of electricity? (Very worried, worried, not too worried, not at all worried)

[General evaluation]

Q17. For Taiwan as a whole, do you believe that the government's promotion of wind power and solar power has a positive or negative impact? (Very good, good, bad, very bad)

[Response: Turnout intention]

Q20. Do you plan to vote in the nuclear power plant referendum scheduled for August 28, 2021?

[Response: Vote choice]

Q21. Do you plan to vote yes or no in the nuclear power plant referendum?

[Partisanship]

Do you identify with any political party? The Kuomintang, People's First Party, and New Party are coded as "Pan-Blue," while the Democratic Progressive Party, Taiwan Statebuilding Party, and New Power Party are coded as "Pan-Green."

Table A1. Descriptive statistics of variables.

	Mean	S.D.	Min	Max
Turnout	0.715	0.452	0	1
Vote for Nuclear Power	0.663	0.473	0	1
Worry	2.701	1.026	1	4
Negative Evaluation on Green Power Policy	2.083	0.835	1	4
Pan-Blue	0.174	0.380	0	1
Pan-Green	0.223	0.417	0	1
Gender	0.496	0.500	0	1
Age	3.381	1.402	1	5
Education	3.730	1.321	1	5
Taiwanese identity	0.572	0.495	0	1
Managers	0.431	0.495	0	1
Office workers	0.223	0.416	0	1
Farmers and fishermen	0.051	0.220	0	1
Workers	0.254	0.435	0	1
Taipei Metropolitan Area	0.178	0.382	0	1
New Taipei and Keelung(North)	0.075	0.264	0	1
Tauyuan, Hsinchu, and Miaoli(Central north)	0.146	0.353	0	1
Taichung, Chungwa, and Nantou(Central)	0.212	0.409	0	1
Yunlin, Chiayi, and Tainan(Central south)	0.197	0.398	0	1
Kaohsiung, Pingtung, and Penghu(South)	0.151	0.358	0	1
Yilan, Hualien, and Taitung(East)	0.038	0.191	0	1